National generation in DEM

Integration of hourly national generation profiles (e.g. from OREES) into the District Energy Model: method and implementation

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Summary

DEM can now take the output of a national energy system model as input: for each national generator (e.g. nuclear, hydro, alpine PV, agri PV, wind, gas plant), an hourly profile of the generation allocated to the municipality. Which generators are used is set in the configuration; any number of generators with any names is possible. DEM then optimises (or simulates) the local system, e.g. heating systems and rooftop PV, with the national generation as given supply.

- Each national profile is an **upper bound**: in each hour the municipality can use up to its allocated share. The unused part is reported as **curtailment**.
- National generation **cannot be exported**. In the optimisation, a binary constraint prevents using national generation and exporting electricity in the same hour, for any objective (costs, emissions, Pareto front).
- National generation is used **before cross-border import**: electricity is only imported from the grid in hours in which all available national generation is used. The remaining demand is supplied by the grid (residual import, with the tariff and CO_2 intensity of `grid_supply`).
- In the optimisation, the optimiser can choose which national generators to use (e.g. cheaper or cleaner ones), as long as no electricity is imported from the grid in that hour.
- If a national generator contains local generation that DEM also models (e.g. national hydro and local `hydro_power`), the local technology must be switched off. Otherwise DEM stops with an error that explains what to change.
- The feature is switched off by default (`national_generation: deployment: false`). Without it, DEM behaves as before.

1 Method

1.1 Previous approach (still used when the feature is off)

Without national generation, the grid import of the municipality is split ex post by the hourly shares of the national electricity mix (file `data/electricity/electricity_mix.feather`): hydro, nuclear, wind, biomass, other and cross-border import. Only the shares are used, not the absolute values, and the split does not influence the optimisation. The grid is an unlimited supply with a tariff and a CO_2 intensity.

1.2 Input: allocated national generation

The national model provides hourly generation profiles per national generator. In a pre-processing step outside DEM, a share is allocated to each municipality. DEM reads, for each generator g, the hourly energy $A_g(t)$ [kWh] available to the municipality. How the allocation is done (e.g. in proportion to electricity demand) is a decision of the pre-processing and is not part of DEM.

Per generator, the configuration also sets a cost [CHF/kWh] and a CO_2 intensity [kgCO_2/kWh] of the used energy, and the local technologies that are contained in the profile (local counterparts, see 1.6).

1.3 Electricity balance

With national generation, the electricity balance of the municipality in each hour t is:

$$D(t) + E(t) = L(t) + \Sigma_g P_g(t) + M(t)$$

D: electricity consumption (incl. heat pumps, EVs, storage charging); E: export; L: local generation (incl. storage discharging); P_g: used national generation of generator g with $0 \le P_g(t) \le A_g(t)$; M: residual grid import.

The residual grid import M is no longer split by the national electricity mix. It is the import not covered by the national generators and is reported as cross-border import. The curtailment of generator g is $A_g(t) - P_g(t)$.

Two rules apply in every hour, in the simulation and in the optimisation:

- **No export of national generation:** national generation is not used in hours with export ($P_g(t) > 0 \Rightarrow E(t) = 0$).
- **National generation before cross-border import:** the grid only supplies electricity if all national generation is used ($M(t) > 0 \Rightarrow P_g(t) = A_g(t)$ for all g).

1.4 Simulation (base scenario and manual scenarios)

Without optimisation, DEM allocates generation to demand by a fixed order. National generation is used after all local generation and before the grid:

- Local generators cover the demand first (unchanged). This gives the remaining need $N(t)$ and the local export.
- National generation covers the remaining need. If more national generation is available than needed, every generator delivers the same fraction of its availability (proportional dispatch):

  $$f(t) = \min(1, N(t) / \Sigma_g A_g(t)), \quad P_g(t) = f(t) \cdot A_g(t)$$
- The grid supplies the rest: $M(t) = N(t) - \Sigma_g P_g(t)$.

Since national generation only covers the remaining need and the grid only supplies what national generation cannot cover, both rules of 1.3 hold by construction in the simulation. The national profiles are applied in the base scenario (as-is) and in all manual scenarios, i.e. every time the local electricity mix is recomputed.

1.5 Optimisation

Each national generator is a supply technology in the Calliope model:

- The hourly resource is the allocated profile $A_g(t)$. The resource is not forced, i.e. the optimiser can use less (curtailment).
- The capacity is fixed to the profile peak; there are no investment or fixed costs.
- Variable cost: cost per kWh used (monetary) and CO_2 intensity per kWh used (emissions). Both enter the objective with the usual weights.
- The generators are located at the grid connection node, like the grid supply. National generation therefore reaches the municipality through the grid connection; if a grid connection capacity is set (`grid_supply: kW_max`), it limits national generation and residual import together.
- National generators have no export carrier.

No re-export of national generation. Because electricity is interchangeable, the optimiser could cover local demand with national generation and export local generation at the same time. This is effectively an export of national generation, and it would be chosen whenever the export revenue or export CO_2 credit is higher than the cost or emissions of national generation. To prevent this, a binary variable $b(t) \in \{0, 1\}$ is added for every hour with available national generation:

$$\Sigma_g P_g(t) \le A(t) \cdot b(t)$$
$$E(t) \le M_E \cdot (1 - b(t))$$

$A(t) = \Sigma_g A_g(t)$; $E(t)$ is the total electricity export (grid export and technology-level export of electricity); M_E is an upper bound of the hourly export (big-M). In each hour, either national generation is used or electricity is exported, not both.

National generation before cross-border import. Without a further constraint, the optimiser would leave national generation unused and import from the grid whenever the grid is cheaper or cleaner. A second binary variable $z(t) \in \{0, 1\}$ is added for every hour with available national generation:

$$\Sigma_g P_g(t) \geq A(t) \cdot z(t)$$
$$M(t) \leq M_G \cdot z(t)$$

$M(t)$ is the grid import (`grid_supply`); M_G is an upper bound of the hourly grid import (big-M). Since $P_g(t) \leq A_g(t)$ for each generator, the first constraint with $z(t) = 1$ means that every generator is fully used. Grid import is only possible with $z(t) = 1$.

Together, the two binaries leave three possible states per hour:

State	National generation	Export	Grid import
1 Export hour ($b = 0$, $z = 0$)	not used	allowed	none
2 National partly used ($b = 1$, $z = 0$)	any amount ≤ available, optimiser chooses the generators	none	none
3 National fully used ($b = 1$, $z = 1$)	all generators fully used	none	allowed

The optimiser can therefore choose which national generators to use only in hours in which national generation covers the remaining demand (state 2). As soon as the grid is needed, the full national mix is used (state 3).

Both constraints are added in every hour with available national generation, independent of prices and emission factors. They therefore hold for cost optimisation, emission optimisation, weighted objectives and the epsilon-constraint Pareto front alike. They turn the optimisation into a mixed-integer problem (two binaries per hour), which increases the run time.

Run time. The national-before-import constraint is easy to solve if national generation is cheaper (or cleaner, depending on the objective) than grid import: the optimiser then prefers it anyway and the constraint is rarely binding. It is hard if national generation is more expensive or emits more than grid import in many hours (e.g. a large gas share in a CO_2 optimisation): the optimiser then tries to avoid it, and the mixed-integer problem has a weak relaxation. In a test with national generation at 0.50 CHF/kWh and a grid tariff of 0.27 CHF/kWh (3 days), Gurobi found the optimum in 114 s, while the open-source solver CBC had not closed the gap below 6 % after 15 minutes. Gurobi is therefore recommended; if needed, increase `solver_option_MIPGap`. In the other tests, Gurobi needed 5–10 s.

Big-M. With `export_big_M_kWh: 'auto'`, M_E is set to twice the sum of the peak values of electricity demand, heat demand, available national generation and the local electricity generation potentials (rooftop PV, alpine PV, wind, hydro). For M_G (`import_big_M_kWh: 'auto'`), the grid connection capacity `grid_supply: kW_max` is used if it is finite, otherwise twice the sum of the peak electricity and heat demand (the grid import cannot exceed the electricity consumption, and the electricity used for heating is at most the heat generated). A tight bound avoids numerical issues (with the integrality tolerance of the solver, a loose bound could allow small violations). If the export or the grid import reaches an estimated bound in any hour, DEM stops with an error, because the bound may then restrict the result; the bound can then be set manually. After the optimisation, DEM also checks that no hour has both national use and export, or both grid import and curtailed national generation, and prints a warning otherwise.

1.6 Avoiding double counting with local technologies

A national profile may already contain generation that DEM models locally: national hydro contains the local hydro power plants, national wind the installed local wind turbines, and national alpine PV the alpine PV that DEM could build locally. For each generator, the configuration lists these local counterparts. Supported counterparts are `hydro_power`, `wind_power` and `solar_pvalpine`.

- If a national generator is active and one of its local counterparts is switched on (`deployment: true`), DEM stops with an error that names the conflicting generator and technology and explains how to resolve it: switch off the local technology or remove the national generator, then rerun. DEM does not switch anything off automatically.
- In the optimisation, switched-off technologies are removed from the model (as before).
- In the simulation, DEM normally keeps the existing generation of a switched-off technology. For local counterparts, the existing generation is set to zero (installed wind power capacity and generation, local hydro

generation), so that it is not counted twice. Manual scenarios that require these technologies still run (with zero generation).

- The scenario `nuclear_phaseout` cannot be combined with national generation (error): the phase-out must be part of the national profiles.

1.7 Costs and emissions

- Used national generation is charged with its cost per kWh; unused (curtailed) generation has no cost. There are no investment, maintenance or capacity costs.
- CO_2 emissions of national generation are the used energy times the CO_2 intensity of the generator. They are included in the total emissions.
- The residual grid import keeps the tariff and CO_2 intensity of `grid_supply` (constant or hourly time series).

1.8 Custom districts

For custom districts spanning several municipalities, the profiles of all municipalities are added, each weighted with the share of the municipality belonging to the district (same weighting as for wind power and EV profiles). If a municipality has no column in a profile file, its profile is set to zero and a warning is printed.

1.9 Assumptions and limitations

- The national model has already dispatched dispatchable plants (gas plant, storage hydro). DEM treats their allocated profiles as fixed upper bounds and does not re-dispatch them.
- **Storage is allowed to shift energy between hours (by design).** Both rules act per hour. National generation can be stored (e.g. in a battery) and exported in a later hour without national use; and grid electricity can be imported and stored in an hour in which all national generation is used, and discharged in a later hour instead of using national generation. Example from the CO_2 test: the battery was charged from the grid before the hours with gas generation, so that the gas share was not used. The origin of stored energy is not tracked.
- Cost and CO_2 intensity are constant per generator (no time series).
- The national profiles are also applied in the base scenario (as-is). The base scenario therefore reflects the national system of the profiles (e.g. a future OREES scenario), not the historic electricity mix.
- The simulation uses a proportional dispatch among national generators; the optimisation uses the generator costs and emissions (cheapest first). Results of the two can therefore differ in the split between generators.
- Because national generation comes before cross-border import, national generation with high costs or emissions (e.g. a gas plant) cannot be replaced by grid import. With a CO_2 objective, it can only be avoided by local generation, flexibility or by reducing demand.
- If several national generators have the same cost and CO_2 intensity, the split of the used energy between them in hours of state 2 is not unique (the totals are). Differentiate the costs if the split matters.
- The time series must match the weather year used in DEM (8760 hours, no leap day).

2 Input format and configuration

2.1 Profile files

- One file per generator, in one directory (`data_dir`).
- Format: `.feather` (recommended) or `.csv`.
- 8760 rows (hours of the year, in the order of the DEM time series; no time column needed).
- One column per municipality, named with the BFS municipality number (e.g. `"2762"`). Other columns are ignored.
- Values: allocated generation in kWh per hour (absolute values, i.e. after allocation to the municipality); no negative values or NaN.

```
data/national_generation/OREES_2050/
```

```
nuclear.feather      hour | 1 | 2 | ... | 2762 | ... | 6810
hydro.feather        -----+---+---+-----+------+-----+-----
alpine_pv.feather       0 |   |   |     | 9876 |     |
agri_pv.feather       ... |   |   |     |  ... |     |
wind.feather         8759 |   |   |     | 9123 |     |
gas.feather          (values in kWh per hour)
```

2.2 Configuration

Section `national_generation` in `inputs.py` (defaults), in a YAML file `national_generation.yaml` or in `technologies.yaml`, or in the `config_dict` of the Python API. An example is given in `config/config_files/EXAMPLE_national_generation.yaml`.

Key	Default	Description
`deployment`	`False`	Switch national generation on/off.
`data_dir`	`''`	Directory of the profile files. '' = `<root_dir>/data/national_generation/`; relative paths are relative to `root_dir`.
`export_big_M_kWh`	`'auto'`	Upper bound of the hourly export in the no-export constraint [kWh]: 'auto' or a number (see 1.5).
`import_big_M_kWh`	`'auto'`	Upper bound of the hourly grid import in the national-before-import constraint [kWh]: 'auto' (grid connection capacity if finite, otherwise estimate) or a number (see 1.5).
`generators`	`{}`	One entry per generator. The name is free (lower-case letters, digits, underscores), e.g. `alpine_pv`. Only listed generators are used.
`file`	-	File name of the profile in `data_dir`.
`cost_CHFpkWh`	-	Cost of used generation [CHF/kWh], ≥ 0.
`co2_intensity`	-	CO_2 intensity of used generation [kgCO_2/kWh], ≥ 0.
`local_counterparts`	-	Local technologies contained in the profile, e.g. `[hydro_power]` or `[]`. They must be switched off.

All four generator keys are required; unknown keys raise an error. Example for the OREES study (local counterparts switched off):

```
national_generation:
  deployment: true
  data_dir: data/national_generation/OREES_2050/
  generators:
    nuclear:   {file: nuclear.feather,   cost_CHFpkWh: 0.0, co2_intensity: 0.0,
                local_counterparts: []}
    hydro:     {file: hydro.feather,     cost_CHFpkWh: 0.0, co2_intensity: 0.0,
                local_counterparts: [hydro_power]}
    wind:      {file: wind.feather,      cost_CHFpkWh: 0.0, co2_intensity: 0.0,
                local_counterparts: [wind_power]}
    alpine_pv: {file: alpine_pv.feather, cost_CHFpkWh: 0.0, co2_intensity: 0.0,
                local_counterparts: [solar_pvalpine]}
    agri_pv:   {file: agri_pv.feather,   cost_CHFpkWh: 0.0, co2_intensity: 0.0,
                local_counterparts: []}
    gas:       {file: gas.feather,       cost_CHFpkWh: 0.0, co2_intensity: 0.0,
                local_counterparts: []}
  hydro_power:    {deployment: false}
  wind_power:     {deployment: false}
  solar_pvalpine: {deployment: false}
```

The cost and CO_2 values above are placeholders. The cost of national generation relative to the grid tariff and the rooftop PV costs strongly influences which local technologies the optimiser builds; it should be set deliberately for the study.

3 Results

The hourly results (`df_base`, `df_scen`, `hourly_results.csv`) contain the following columns ([kWh] or [kgCO_2]); annual sums are in the annual results with the suffix `_yr`.

Column	Description
`v_e_nat_<g>`	Used national generation of generator <g>
`v_e_nat_pot_<g>`	Available (allocated) national generation of <g>
`v_e_nat_curt_<g>`	Curtailed (unused) national generation of <g>
`v_co2_nat_<g>`	CO_2 emissions of <g> (included in `v_co2_tot`)
`v_e_nat`	`v_e_nat_pot`, `v_e_nat_curt`: totals over all generators
`m_e` = `m_e_cbimport`	Residual grid import (not covered by local or national generation)
`m_e_ch, m_e_ch_hydro, ...`	Split of the import by the national mix: zero when national generation is active

The electricity balance plots (hourly, daily, weekly) show one series per national generator ("National <g>"), and the Sankey diagram shows them as electricity inputs. The cost module reports the energy costs of national generation in the monetary breakdown (`national_generation`) and its emissions in the CO_2 breakdown (`emissions_national_generation`).

4 Implementation: where it is in the code

All paths are relative to `src/district_energy_model/`.

File	Function / location	What it does
`techs/dem_tech_national_generation.py`	(new module)	All national generation logic.
	`check_national_generation_config`	Validates the configuration; raises the errors for conflicts with local counterparts and with `nuclear_phaseout`.
	`validate_tech_dict`	Checks generator names, required/unknown keys, values, supported counterparts, `export_big_M_kWh`.
	`is_deployed`, `get_local_counterparts`	Helpers used by `dem.py` (also work with input files without the new section).
	`resolve_data_dir`, `load_profile`	Locate and read profile files; select the municipality column; weight columns for custom districts; checks (8760 rows, no NaN, ≥ 0).
	`class NationalGeneration`	Tech class (child of `TechCore`): profiles, used generation per generator, costs, CO_2.
	`NationalGeneration.dispatch`	Proportional dispatch in the simulation (1.4).
	`NationalGeneration.create_techs_dict`	Calliope supply techs `national_<g>` (1.5).
	`NationalGeneration.update_df_results`	Result columns (section 3).
`input_files/inputs.py`	`scen_techs['national_generation']`	Default configuration (switched off, no generators).
`dem_paths.py`	`national_generation_dir`	Default profile directory.
`dem_techs.py`	imports	Registers `NationalGeneration`.
`dem.py`	`DistrictEnergyModel.__init__`	Calls the configuration check; stores the local counterparts.
	`__generate_base_scenario`	Creates the tech instance (after `grid_export`); sets the existing generation of local counterparts to zero; uses `compute_base_grid_import_without_split` instead of the mix split.
	`generate_scenario`	Note for counterparts kept with zero generation; passes `replaced_techs` to the scenario checks.

File	Function / location	What it does
dem_energy_balance.py	get_local_electricity_mix	After local generation: national dispatch, then residual grid import. Used by the base scenario and all manual scenarios.
	electricity_balance_test	National generation added to the balance; new checks: used ≤ available; no grid import while national generation is curtailed.
techs/dem_tech_grid_supply.py	compute_base_grid_import_without_split	Import without mix split (m_e_ch_* = 0, m_e_cbimport = m_e).
techs/dem_tech_wind_power.py	remove_existing_generation	Removes installed wind power (capacity and generation) if wind is a counterpart.
dem_calliope.py	CalliopeOptimiser.__init__	Gets the tech instance.
	run_optimisation	Adds profiles to the time series (national_generation); enables custom constraints; calls the no-export constraint.
	__create_techs_dict	Adds the national techs to the grid connection node.
	get_optimal_output_df	Reads the used national generation from the results; calls the export check.
	__estimate_national_big_M	Big-M estimate for 'auto' (1.5).
	__get_national_export_big_M, __get_national_import_big_M	Big-M of export and grid import (import: grid connection capacity if finite).
	__check_national_generation_results	Error if export or grid import reaches an estimated big-M; warning if national use and export, or grid import and curtailment, occur in the same hour.
dem_calliope_cc.py	national_generation_no_export_constraints	Binary *b(t)* and the no-export constraints (1.5).
	national_generation_before_import_constraints	Binary *z(t)* and the national-before-import constraints (1.5).
	get_electricity_export_flows	All electricity export flows (tech-level export of electricity/wp_electricity and grid export).
dem_costs.py	get_old_capacities	National generation excluded from capacity and CAPEX/OPEX calculations (also in get_replacement_capacities, calculate_total_annual_costs).
	calculate_CO2_emissions	Adds emissions_national_generation.
dem_helper.py	check_tech_for_scenario	New argument replaced_techs: counterparts do not block manual scenarios.
	add_missing_keys	v_e_nat, v_e_nat_pot added (zero if not deployed).
dem_output.py	get_electricity_balance_series	Adds national series to the electricity balance plots.
	plot_sankey_total	National generators as electricity inputs.

Further files: config/config_files/EXAMPLE_national_generation.yaml (example configuration), CHANGELOG.md.

4.1 Program flow

Step	With national generation
Model initialisation	Configuration check (errors for conflicts). Profiles are read for the municipality or custom district (8760 h).
Base scenario	Local counterparts set to zero. Local mix: local generation, then national generation (proportional), then grid. Import not split by the mix.
Scenario: time reduction	Profiles cut to number_of_days.

Step	With national generation
Manual scenarios	Every recomputation of the local mix dispatches national generation again.
Optimisation	National techs at the grid connection node; binary no-export and national-before-import constraints; results read back; checks.
Costs / CO_2	Energy costs and emissions of used national generation.

5 Tests

The feature was tested with Allschwil (2762), 3 days, synthetic profiles for six generators (nuclear, hydro, wind, alpine PV, agri PV, gas). The optimisation cases were run with Gurobi 12.0.3 and with the open-source solver CBC 2.10. Checks in each case: used ≤ available, curtailment = available – used, old mix split zero, residual import = cross-border import, local counterparts without generation, no hour with both national use and export, no hour with both grid import and curtailment, proportional dispatch in the simulation, DEM energy balance tests.

Test case	Description	CBC	Gurobi
sim_reference, opt_reference	Feature off (simulation, optimisation). Results identical to the code before the change (same costs and CO_2).	pass	pass (5 s)
sim_national	Simulation (base scenario 8760 h, scenario 72 h). Proportional dispatch, curtailment, counterparts zero, no export of national generation, no grid import while national generation is curtailed.	pass	n/a
sim_national_scenarios	Simulation with manual scenarios pv_integration and fossil_heater_retrofit (hydro_power, wind_power switched off).	pass	n/a
opt_national	Optimisation, cost objective. Curtailment only in hours in which local PV covers the demand; costs of the cost module equal the Calliope objective.	pass	pass (6 s)
opt_national_co2	Optimisation, CO_2 objective. All renewable national generation used. Gas is not used, and no grid import occurs in the gas hours (the battery covers them, see 1.9).	pass	pass (5 s)
opt_national_reexport	Stress test: national cost 0, feed-in tariff 0.20 CHF/kWh. With the constraint: no hour with national use and export. Without the no-export constraint (counterfactual run, CBC): 76.7 MWh exported in 10 hours, and the post-solve check warns.	pass	pass (5 s)
opt_national_expensive	Stress test: national cost 0.50 CHF/kWh, grid tariff 0.27 CHF/kWh. No hour with grid import while national generation is curtailed. CBC: only with a MIP gap of 10 % (default gap not reached within 15 min). Without the national-before-import constraint (counterfactual run, CBC): up to 12.4 MWh per hour imported while national generation is curtailed; the post-solve check warns and the energy balance test (13) stops the run.	pass (gap 10 %)	pass, optimal (114 s)
opt_national_subset	Only nuclear and gas; local hydro and wind remain switched on (no counterparts).	pass	pass (10 s)
opt_national_scaled	calliope_energy_scaling_factor = 1000: same costs, emissions and totals as unscaled.	pass	not possible*
plots / results files	Electricity balance plots (hourly, daily, weekly) and Sankey contain the national generators; hourly results contain the new columns; cost breakdown contains national generation.	pass	n/a
err_conflict	National hydro/wind/alpine PV with local counterparts switched on: error listing the conflicts and how to resolve them.	pass	n/a
err_nuclear_phaseout	Combination with nuclear_phaseout: error.	pass	n/a
err_unknown_key	Unknown generator key (must_take): error.	pass	n/a
err_missing_file	Missing profile file: error with path.	pass	n/a

* Existing issue, independent of national generation: DEM passes FeasibilityTol / scaling factor (1e-6 / 1000) to Gurobi, which rejects it as below its minimum of 1e-9 (`dem_calliope.py`, `__create_run_dict`).

Not tested: custom districts, the Pareto front, full-year runs (run time with two binary variables per hour).